

CSCI 5244: Homework Set 3

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1. Given two vectors $x, y \in \mathbb{R}^n$, using amplitude encoding we can encode them as quantum states

$$|x\rangle = \sum_{k=1}^n \frac{x_k}{\|x\|} |k\rangle, \quad |y\rangle = \sum_{k=1}^n \frac{y_k}{\|y\|} |k\rangle$$

where $\|\cdot\|$ denotes the Euclidean norm and $\{|1\rangle, |2\rangle, \dots, |n\rangle\}$ are orthonormal quantum states.

- (a) Show that the inner product $x^T y = \|x\| \|y\| \langle x|y\rangle$

Begin solution

Rewrite the given amplitudes to reflect $\langle x|y\rangle$

$$\langle x| = \sum_{j=1}^n \frac{x_j}{\|x\|} \langle j|, \quad |y\rangle = \sum_{k=1}^n \frac{y_k}{\|y\|} |k\rangle$$

$\langle x|$ is the conjugate transpose, so it's amplitudes are $(x_j/\|x\|)^*$. The conjugate drops out because we are in the $\mathbb{R}^n$ space, and don't have to deal with complex numbers.

Now we can write $\langle x|y\rangle$ as the double sum

$$\langle x|y\rangle = \sum_{j=1}^n \sum_{k=1}^n \frac{x_j}{\|x\|} \frac{y_k}{\|y\|} \langle j|k\rangle$$

Since these are orthonormal quantum states for every pair (j, k) where $j \neq k$ results in a 0, else $j = k$ results in a 1. This is just the Kronecker Delta δ_{jk} , thus the sum reduces to

$$\langle x|y\rangle = \sum_i \frac{x_i y_i}{\|x\| \|y\|}$$

Thus we get

$$\langle x|y\rangle = \frac{1}{\|x\| \|y\|} \sum_{i=1}^n x_i y_i \Rightarrow \|x\| \|y\| \langle x|y\rangle = \sum_{i=1}^n x_i y_i = x^T y$$

- (b) The outer product xy^T is an $n \times n$ matrix. How to encode xy^T quantumly using $|x\rangle$ and $|y\rangle$?

Begin solution

xy^T is an $n \times n$ matrix with (j, k) entry $x_j y_k$ which is proportional to $|x\rangle \langle y|$, the outer product.

Expanding the outer product with given amplitudes gives us

$$|x\rangle = \sum_{j=1}^n \frac{x_j}{\|x\|} |j\rangle, \langle y| = \sum_{k=1}^n \frac{y_k}{\|y\|} \langle k|$$

Writing as a double sum

$$|x\rangle \langle y| = \frac{1}{\|x\|\|y\|} \sum_{j=1}^n \sum_{k=1}^n x_j y_k |j\rangle \langle k|$$

This differs from $x_j y_k$ by the same constant $\frac{1}{\|x\|\|y\|}$ for every entry, so

$$A = \|x\|\|y\| |x\rangle \langle y| = \sum_{j=1}^n \sum_{k=1}^n x_j y_k |j\rangle \langle k|$$

(c) Suppose the answer to (b) is A . Show that the inner product $x^T y = \text{tr } A$.

Begin solution

$\text{tr}(A) = \sum_l \langle l|A|l\rangle$, the sum of the diagonal entries.

From 1b, we can substitute A to be

$$\sum_j \sum_k x_j y_k |j\rangle \langle k|$$

Applying the trace

$$\begin{aligned} \text{tr}(A) &= \sum_l \sum_j \sum_k \langle l| (x_j y_k |j\rangle \langle k|) |l\rangle \\ &= \sum_l \sum_j \sum_k x_j y_k \langle l|j\rangle \langle k|l\rangle \\ &= \sum_l \sum_j \sum_k x_j y_k \langle k|l\rangle \langle l|j\rangle \end{aligned}$$

note: also assume that all indexes be it j, k, l, g all start at 1. I just don't want to write it out every time.

The brackets can be rearranged because we are dealing with real numbers.

Because $\sum_l |l\rangle \langle l| = I$ we can replace that in the equation

$$\begin{aligned} &= \sum_j \sum_k x_j y_k \langle k|I|j\rangle \\ &= \sum_j \sum_k x_j y_k \langle k|j\rangle \end{aligned}$$

Apply the Kronecker delta δ_{kj} to get

$$\sum_j x_j y_j = x^T y = \text{tr}(A)$$

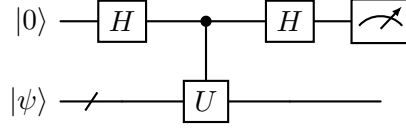
Thus it is proven that the inner product $x^T y = \text{tr}(A)$

- (d) Design a quantum algorithm to determine the sign of $x^T y$ i.e. if it is positive or negative.

Begin solution

Because the encoding requires $x, y \neq 0$, the norm has to be $\|x\|\|y\| > 0$ for $x^T y = \|x\|\|y\| \langle x|y \rangle$. Thus, it does not affect the sign of $\langle x|y \rangle$, thus $\text{sign}(x^T y) = \text{sign}(\langle x|y \rangle)$.

Starting to build the algorithm, we can take inspiration from the [Hadamard test](#) which has the following circuit



We assume we have access to the state-preparation unitary operators U_x and U_y for $|x\rangle$ and $|y\rangle$ respectively giving us $|x\rangle = U_x |0\rangle$ and $|y\rangle = U_y |0\rangle$ since any unit vector extends to an orthonormal basis.

Writing the conjugate transpose of $|x\rangle$ to get $\langle x| = \langle 0| U_x^\dagger$.

Thus $\langle x|y \rangle = \langle 0| U_x^\dagger U_y |0\rangle$ which matches the Hadamard test $|\psi\rangle = |0\rangle$ and $U = U_x^\dagger U_y$.

Following the design of the Hadamard test for the real expression (since we are not dealing with imaginary numbers right now), the steps are as follows

First, we prepare an ancilla qubit $|0\rangle$ and data register $|0\rangle$ and apply a Hadamard gate to the ancilla qubit

$$\begin{aligned} |0\rangle^A, |0\rangle^D &\xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle \\ &\xrightarrow{c-U_y} \frac{1}{\sqrt{2}}(|0\rangle |0\rangle + |1\rangle U_y |0\rangle) \\ &\xrightarrow{c-U_x^\dagger} \frac{1}{\sqrt{2}}(|0\rangle |0\rangle + |1\rangle U_x^\dagger U_y |0\rangle) \\ &\xrightarrow{H} \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle + \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes U_x^\dagger U_y |0\rangle \right) \\ &= \frac{1}{2} \left((|0\rangle + |1\rangle) |0\rangle + (|0\rangle - |1\rangle) U_x^\dagger U_y |0\rangle \right) \\ &= \frac{1}{2} \left(|0\rangle (|0\rangle + U_x^\dagger U_y |0\rangle) + |1\rangle (|0\rangle - U_x^\dagger U_y |0\rangle) \right) \end{aligned}$$

The probability of measuring the qubit 0 to be in state $|0\rangle$ is (using $U = U_x^\dagger U_y$)

$$\begin{aligned} P(0) &= \left\| \frac{1}{2}(|0\rangle + U |0\rangle) \right\|^2 = \frac{1}{4} (\langle 0| + \langle 0| U^\dagger) (|0\rangle + U |0\rangle) \\ &= \frac{1}{4} \left[\langle 0|0\rangle + \langle 0|U|0\rangle + \langle 0|U^\dagger|0\rangle + \langle 0|U^\dagger U|0\rangle \right] \end{aligned}$$

Note that $\langle 0|0\rangle = 1$, $\langle 0|UU^\dagger|0\rangle = 1$, and $\langle 0|U^\dagger|0\rangle = (\langle 0|U|0\rangle)^*$, and $z + z^* = 2\text{Re}(z)$ so we get

$$\begin{aligned}
&= \frac{1}{4}(2\text{Re}(\langle 0|U|0\rangle) + 2) \\
&= \frac{1}{2}(1 + \text{Re}(\langle x|y\rangle))
\end{aligned}$$

The squared norm of the data state

$$P(0) = \frac{1}{2}(1 + \text{Re} \langle 0|U_x^\dagger U_y|0\rangle) = \frac{1}{2}(1 + \text{Re} \langle x|y\rangle)$$

Since $\langle x|y\rangle$ is already real, Re changes nothing.

The circuit is then repeated N times to estimate $P(0)$. The ancilla is measured in the computational basis and $P(0) \approx N_0/N$. The sign is measured to be positive if $P(0) > 1/2$ and negative if $P(0) < 1/2$. If $P(0) = 1/2$ it means $\langle x|y\rangle = 0$. One thing to note: the closer $\langle x|y\rangle$ is to 0, the larger N must be to tell the two sides apart.

2. Given N vectors $x_1, x_2, \dots, x_N \in \mathbb{R}^n$. Define an N -by- N matrix K with entry $K_{ij} = x_i^T x_j$. Matrix K can be encoded as a quantum state described by the density matrix $\rho = \sum_{i=1}^N \sum_{j=1}^N K_{ij} |i\rangle \langle j|$ where $\{|1\rangle, |2\rangle, \dots, |N\rangle\}$ are orthonormal quantum states. Explain in detail how you would prepare state ρ .
3. To be submitted via email.